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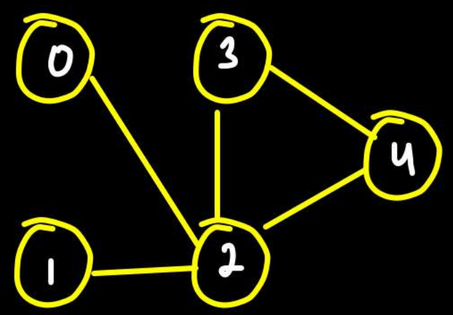
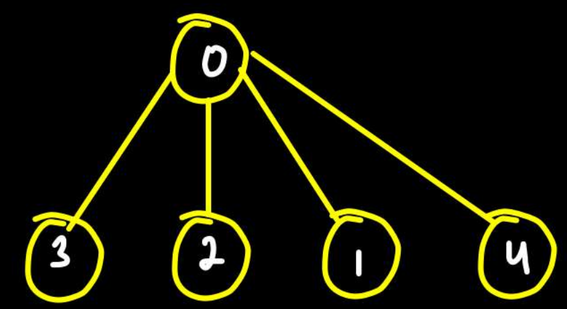
Undirected Graph Cycle (DSU)





0,2

1,2

2,3

2,4

3,4

if (u,v) are not in same component → union

else → cycle found

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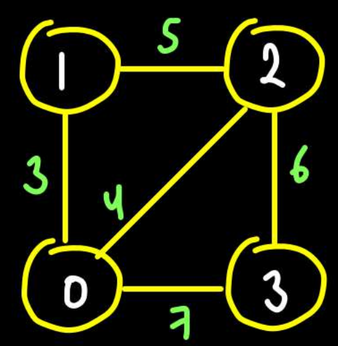
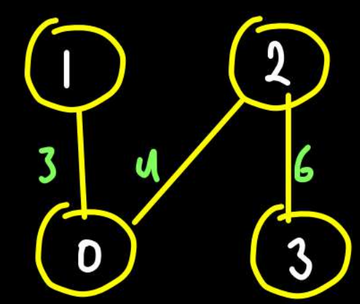
Kruskal's Algorithm

MST





Steps

1) sort the edge list w.r.t wt.

2) If u & v are disjoint then union them & take the edge.

✓ 0, 1, 3

✓ 0, 2, 4

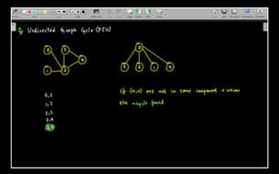
✗ 1, 2, 5

✓ 2, 3, 6

✗ 0, 3, 7

Sorted Edge List

mstSum = 3 + 4



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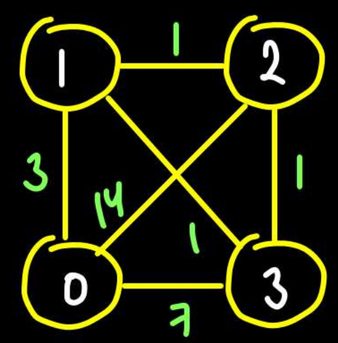
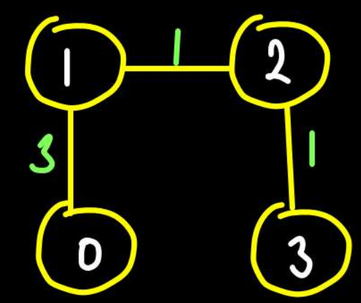
Kruskal's Algorithm

MST





Steps

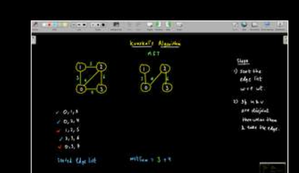
1) sort the edge list w.r.t wt.

2) If u & v are disjoint then union them & take the edge.

✓ 1, 2, 1
✓ 2, 3, 1
✗ 1, 3, 1
✓ 0, 1, 3
✗ 0, 3, 7
✗ 0, 2, 14

Sorted Edge List

mstSum = 1 + 1 + 3 = 5



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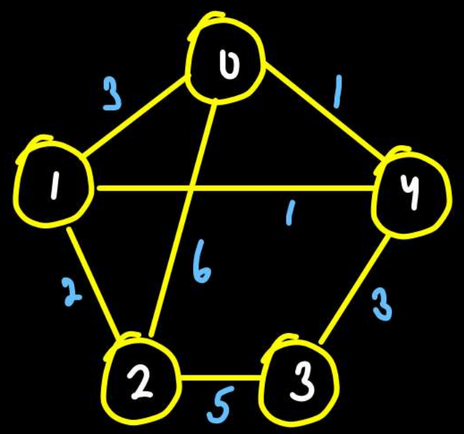
Floyd-Warshall Algorithm




```
for(k=0 to n-1){
    for(u=0 to n-1){
        for(v=0 to n-1){
            dist[u][v] = min(dist[u][v], dist[u][k] + dist[k][v])
```

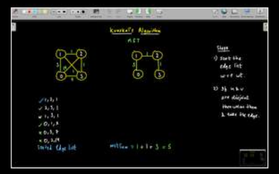
	0	1	2	3	4
0	0	3	6	∞	1
1	3	0	2	∞	1
2	6	2	0	5	∞
3	∞	∞	5	0	3
4	1	1	∞	3	0

dist

1 1 3

1 3

3



Loading scene (24/25)

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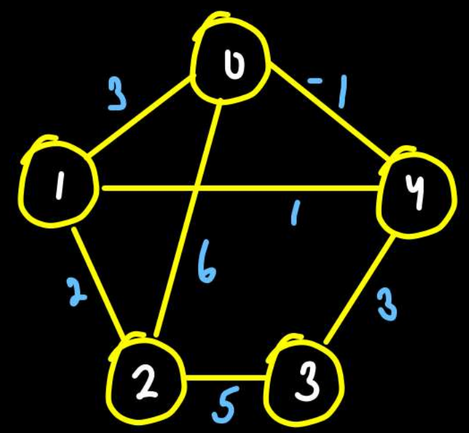
Floyd-Warshall Algorithm (Brute-Force)



$$dist[u][v] = \min(dist[u][v],\ dist[u][k] + dist[k][v])$$

	0	1	2	3	4
0	0	3	6	∞	1
1	3	0	2	∞	1
2	6	2	0	5	∞
3	∞	∞	5	0	3
4	1	1	∞	3	0

dist

	0	1	2	3	4
0	0	2	4	4	1
1	2	0	2	4	1
2	4	2	0	5	3
3	∞	∞	5	0	3
4	1	1	∞	3	0

dist

min-distances

$2 \xrightarrow{\cancel{\infty}\ 3} 4$

$2 \xrightarrow{4} 0 \xrightarrow{1} 4$ ✓

$2 \xrightarrow{2} 1 \xrightarrow{1} 4$ ✓

$2 \longrightarrow 2 \longrightarrow 4$ ✗

$2 \xrightarrow{5} 3 \xrightarrow{3} 4$ ✗

$2 \longrightarrow 4 \longrightarrow 4$ ✗

DIY